CampusNotes - Academic Study Resource

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING - SEMESTER 5

OFFICIAL ACADEMIC STUDY MATERIAL - CSE CURRICULUM

Experiment 10: Travelling Salesperson Problem TSP with Branch & Bound

Course: Design Analysis and Algorithms Laboratory [23CS3505]
Category: Experiments / Practicals | Experiment #10 | Academic Year: 2026-27

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1. Overview & Objective:

Solve the symmetric TSP using reduced cost matrix and Lowest-Cost Branch and Bound search.

2. Practical Tasks & Verification Steps:

1. Initialize cost matrix representing distance between all city pairs.
2. Construct initial reduced matrix subtracting row minima and column minima to establish lower bound.
3. Define search tree nodes tracking current path, visited bitmask, reduced matrix, and cumulative lower bound.
4. Select next exploration branch using priority queue based on smallest lower bound estimate.

Submission Guidelines & Verification:

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